

Cheap *and* Accurate Loss-Landscape Visualisation: Budget-Optimal Probing with a Certified Error

Abstract

Loss-landscape figures are read quantitatively — for the width of a basin, the direction of the gradient field, the sharpness of a minimum — yet are produced without any statement of their own accuracy. We pose the problem as budgeted stochastic function estimation on a 2-plane and derive what a compute budget C , in example-forward-equivalents, can buy. Two facts drive the result. On a 2-plane the exact restricted gradient and the exact 2×2 restricted Hessian cost one backward pass and two Hessian-vector products, independently of model size, so a single probe yields a complete second-order Taylor model. And splitting C between the number of probes and the depth of each has a closed-form optimum: a partition-of-unity Taylor reconstruction is best at $n^* \propto C^{1/4}$ anchors, with error decaying as $C^{-3/8}$ — the nonparametric minimax rate for the smoothness assumed. The same analysis shows that plotting a grid of noisy loss values, which is standard practice, is an *interpolating* estimator whose error cannot fall below the per-point Monte-Carlo noise, and that curvature read off such a grid converges at $C^{-1/8}$ against $C^{-1/4}$ for direct Hessian probing. Every rendered surface ships with a variance-corrected hold-out certificate, unbiased for its own mean squared error and carrying a distribution-free confidence bound. On a 402,986-parameter CNN (CIFAR-10) and a 5,018,112-parameter transformer (WikiText-2), scored against exact full-dataset references, the measured exponents track the predicted ones and the 95% bound holds in 99.6% of runs. Six load-bearing lemmas are machine-checked in Lean 4.

1 Introduction

A loss-landscape figure is a claim about geometry. Readers use them to argue that one minimum is flatter than another, that a barrier separates two solutions, that an optimiser was pushed in a particular direction. Those are quantitative claims, and they are made from pictures that carry no error bar.

The pictures are also expensive. Producing an $n \times n$ grid of loss values costs n^2 passes over data; at $n = 100$ and a dataset of 10^4 examples that is 10^8 example-forward-passes for a single frame. The response in practice is to shrink the per-point sample — a mini-batch instead of the dataset — and this is where the trouble starts, because it trades a cost that is visible for an error that is not.

The question. We think the right question is not *how do I interpolate a sparse grid cheaply*, but:

Given a compute budget C , what is the smallest achievable error of a rendered loss surface, and which sampling and reconstruction strategy achieves it?

Posed this way the problem has a definite answer, and the answer contradicts common practice in three specific ways.

Contributions.

1. **A cost model and an allocation theorem** (Section 5). Under a budget C split as n anchors of B examples each, the error of a partition-of-unity Taylor reconstruction is $c_1 M_3 h^3 + c_2 \sigma B^{-1/2}$ with $h \simeq Rn^{-1/2}$; its minimum is attained at $n^* \propto (C/\kappa)^{1/4}$ with value $\propto (M_3 R^3)^{1/4} (\kappa \sigma^2 / C)^{3/8}$ (Theorem 6). The exponent matches the minimax rate for a C^3 function on a 2-dimensional domain, so the scheme is rate-optimal. *More budget should buy depth, not resolution*: n^* grows only as the fourth root.
2. **Two negative results about current practice**. Plotting a grid of noisy loss estimates is an interpolating estimator, and its error at the grid points equals the noise exactly (Proposition 10); refining the grid at fixed per-point sample size cannot converge. And because reading curvature off values requires second differencing, *sharpness estimated from a value-only grid converges at $C^{-1/8}$* — a $256\times$ budget increase to halve the error — against $C^{-1/4}$ when the Hessian is probed directly (Table 1). We measure both.
3. **Derivative probing is cheap on a plane — and we measure exactly how cheap, including where it does not pay** (Section 3). The restricted gradient is 2 inner products after one backward pass; the restricted Hessian is 2 Hessian-vector products, *independently of N* . Measured multipliers are $\kappa_1 = 3.9$, $\kappa_2 = 20.8$ (CNN) and $\kappa_1 = 3.68$, $\kappa_2 = 28.1$ (transformer). Since $E^* \propto \kappa^{3/8}$, a second-order probe costs about $3\times$ in error on the value surface and buys about $1.8\times$ back through a smaller bias constant — so for the *surface* it is close to a wash, and the first-order probe is often the better buy. We say so, and show the measurement. For the gradient and curvature fields the comparison is not close.
4. **A certificate** (Section 6). Independent hold-out probes give a variance-corrected estimate of the reconstruction’s mean squared error that is unbiased *conditional on the reconstruction*, with an empirical-Bernstein upper bound and a distribution-free quantile statement. It assumes no smoothness and no noise model; a bad reconstruction certifies as bad.
5. **Projection fidelity, measured rather than claimed** (Section 7). We show the Frobenius-optimal *affine* plane is centred at the trajectory mean, not anchored at θ_0 (Proposition 14), and — more importantly — that the captured-variance ratio ρ_2 is the wrong diagnostic twice over: on both subjects the θ_0 -anchored plane reports a *higher* ρ_2 while having a *larger* residual, and neither quantity predicts the loss-space error at all. We instead report the projection gap $\gamma_t = L(\theta_t) - L(\Pi\theta_t)$, measured in loss units against the relief of the surface.
6. **Refining a grid makes the gradient and curvature fields monotonically worse**. Not just slower to converge — worse. Holding the per-point sample fixed and spending extra budget on resolution shrinks h , and the h^{-2} noise amplification of second differencing outruns the shrinking bias: we measure fitted exponents of -0.29 and -0.695 (negative, i.e. error growing with budget) for the gradient and curvature fields. A practitioner who responds to a noisy-looking landscape by refining the grid is making it worse.
7. **An implementation whose speed is part of the method** (Section 8). Fixed per-anchor overhead τ enters the optimum, so reducing it improves attainable *accuracy*, not just wall-clock. Fused CUDA kernels reach 88.8% of peak DRAM bandwidth and cut the reconstruction sweep by $225\times$ over PyTorch; a Triton implementation of all four kernels was written and benchmarked against them and lost on every operation.
8. **Machine-checked foundations** (Section 10). The allocation optimum and its attainment, the partition-of-unity bound, the interpolation floor, the unbiasedness of the certificate, and the

affine-centring decomposition are formalised in Lean 4 with mathlib, depending on no axioms beyond Lean’s three.

What this paper does not claim. We do not claim a new projection method; the plane is chosen by trajectory PCA, as in prior work. We do not claim the $C^{-3/8}$ rate is new to statistics — it is the classical minimax rate, and identifying the visualisation problem with that setting is the contribution. We do not claim the certificate bounds the sup-norm error; it bounds the mean squared error and a quantile, which is what finitely many samples support.

2 Setup: what is being drawn, and what it costs

2.1 The target

Almost all prior work leaves the object being plotted implicit, which makes accuracy unanswerable: if the target is the population risk, no finite computation evaluates it, and there is no ground truth to compare against. We fix it.

Definition 1 (Target). Let $\mathcal{D}$ be a fixed finite evaluation set and $\ell(\theta; z)$ a per-example loss. The *landscape* is the function $\mathcal{L}_{\mathcal{D}}(\theta) = |\mathcal{D}|^{-1} \sum_{z \in \mathcal{D}} \ell(\theta; z)$.

Three things follow. $\mathcal{L}_{\mathcal{D}}$ is computable exactly, so dense reference surfaces are genuine ground truth. A probe that draws B elements of $\mathcal{D}$ uniformly without replacement is unbiased for it, with a variance estimable from the same draw. And cost has a single natural unit.

Definition 2 (Budget). One *example-forward-equivalent* is the marginal cost of one forward pass on one element of $\mathcal{D}$. A budget C is a number of such units.

2.2 The plane, and the restricted loss

Fix an affine plane $\mathcal{S} = c + \text{span}\{v_1, v_2\}$ with $V = [v_1 \ v_2]^\top$ orthonormal, and write

$$\ell(\alpha) = \mathcal{L}_{\mathcal{D}}(c + \alpha_1 v_1 + \alpha_2 v_2), \quad \alpha \in \mathbb{R}^2.$$

Everything drawn is a function of α on a rectangular domain whose choice is fixed by a stated criterion (Section 9); R denotes its effective radius, the geometric mean of the two half-widths, which is the scale the error analysis uses. Note this is a *restriction* of $\mathcal{L}_{\mathcal{D}}$, not a projection of it: structure orthogonal to $\mathcal{S}$ is invisible no matter how the plane is chosen. Section 7 measures what that costs.

2.3 Cost model

A probe at α using B examples and returning derivatives up to order r costs

$$\text{cost}(r, B) = \tau_r + \kappa_r B \tag{1}$$

example-forward-equivalents. κ_r is the marginal per-example multiplier; τ_r is the fixed per-anchor overhead — writing θ into the model, kernel launches, autograd graph construction — expressed in the same unit. Both are *measured* for each subject by timing probes at several B and regressing (Table 2); neither is assumed.

3 Probing a plane

3.1 Second-order information is cheap here

The observation that drives the method:

Proposition 3 (Restricted derivatives). *For ℓ as above, $\nabla\ell(\alpha) = V\nabla\mathcal{L}_{\mathcal{D}}(\theta) \in \mathbb{R}^2$ and $\nabla^2\ell(\alpha) = V\nabla^2\mathcal{L}_{\mathcal{D}}(\theta)V^\top \in \mathbb{R}^{2 \times 2}$. The gradient costs one backward pass and two inner products; the Hessian costs two Hessian-vector products $\nabla^2\mathcal{L}_{\mathcal{D}}v_1$, $\nabla^2\mathcal{L}_{\mathcal{D}}v_2$ [Pearlmutter, 1994] and four inner products. Neither count depends on N .*

So a single probe returns six numbers — a complete second-order Taylor model of ℓ at α — at cost κ_2 times a value. A value-only scheme must recover the same six numbers by differencing six nearby evaluations, which costs more *and* amplifies their noise by $\Theta(h^{-2})$ in the second-order coefficients.

Table 2 gives the measured multipliers. They are not small: κ_2 is 20.8 for the CNN and 28.1 for the transformer, so a second-order probe is worth roughly twenty to thirty value evaluations. Section 5 shows what that buys and Section 9 whether it is worth it — the answer differs by quantity.

3.2 Every probe measures its own noise

A probe returns, from the same draw and at no extra cost: the sample variance of the per-example losses, giving $\widehat{\text{Var}}(\hat{\ell}) = \hat{s}^2 B^{-1}(1 - B/|\mathcal{D}|)$ with the finite-population correction; and, by splitting the draw into two independent halves $\hat{g}_A, \hat{g}_B$, the estimator $\widehat{\text{Var}}(\hat{g}) = (\hat{g}_A - \hat{g}_B)^2/4$, which is unbiased for the variance of the full-sample mean. The same construction gives $\widehat{\text{Var}}(\hat{H})$. These variances are what make the reconstruction certifiable.

4 Reconstruction: partition-of-unity Taylor patches

Attach to each anchor α_i its measured second-order model

$$Q_i(\alpha) = \hat{\ell}_i + \hat{g}_i^\top(\alpha - \alpha_i) + \frac{1}{2}(\alpha - \alpha_i)^\top \hat{H}_i(\alpha - \alpha_i), \quad (2)$$

and blend with compactly supported Wendland C^2 weights $w_i(\alpha) = \varphi(\|\alpha - \alpha_i\|/\rho_i)$, $\varphi(r) = (1 - r)_+^4(4r + 1)$ [Wendland, 1995], normalised to a partition of unity [Shepard, 1968, Babuška and Melenk, 1997]:

$$\mathcal{R}(\alpha) = \frac{\sum_i w_i(\alpha) Q_i(\alpha)}{\sum_i w_i(\alpha)}. \quad (3)$$

Two properties matter.

Lemma 4 (Local to global, no loss). *If $w_i \geq 0$, $\sum_i \hat{w}_i = 1$, and $|\ell - Q_i| \leq \varepsilon$ on the support of w_i for every i , then $|\ell - \mathcal{R}| \leq \varepsilon$ pointwise.*

Proof. $\ell - \mathcal{R} = \sum_i \hat{w}_i(\ell - Q_i)$, so $|\ell - \mathcal{R}| \leq \sum_i \hat{w}_i |\ell - Q_i| \leq \varepsilon \sum_i \hat{w}_i = \varepsilon$. $\square$

The step is elementary and it is exactly the reason to use a partition of unity rather than a scattered-data interpolant: no Lebesgue constant appears, so the global bound is the local bound. Combined with Taylor’s theorem with Lagrange remainder — if the third directional derivative of ℓ is bounded by M_3 on the patch then $|\ell - Q_i| \leq \frac{1}{6} M_3 \rho_i^3$ — Lemma 4 gives the cubic approximation order the allocation rests on. Both are machine-checked (Section 10).

Proposition 5 (Exact quadratic reproduction). *If ℓ is a quadratic then $Q_i = \ell$ for every i , hence $\mathcal{R} = \ell$ identically — for any anchor placement, any radii, and any $n \geq 1$.*

No interpolant of *values* has this property at $n < 6$, and none has it at all when the values are noisy.

The rendered gradient is the gradient of the rendered surface. Because (3) is differentiable in closed form,

$$\nabla \mathcal{R} = \frac{1}{W} \left[\sum_i w_i \nabla Q_i + \sum_i (Q_i - \mathcal{R}) \nabla w_i \right], \quad W = \sum_i w_i, \quad (4)$$

and we render *that*. Pipelines that interpolate a gradient field separately from the values produce a quiver plot that is not the gradient of the surface it is drawn on, and the inconsistency is invisible to the reader. The second term in (4) is what makes the difference; our CUDA kernel computes both terms and we verify (4) against central differences of the rendered surface to 5×10^{-5} relative.

5 How to spend a budget

5.1 The allocation theorem

Write the error of (3) as an approximation term plus a Monte-Carlo term. With n anchors quasi-uniform on a domain of radius R the spacing is $h \simeq Rn^{-1/2}$, so

$$E(n, B) \simeq \underbrace{c_1 M_3 R^3 n^{-3/2}}_{\text{approximation}} + \underbrace{c_2 \sigma B^{-1/2}}_{\text{Monte Carlo}}, \quad n(\tau_r + \kappa_r B) = C. \quad (5)$$

Theorem 6 (Budget allocation). *Let $a = c_1 M_3 R^3 > 0$ and, in the regime $\tau_r \ll C/n$, $B = C/(\kappa_r n)$. Substituting $u = n^{1/2}$, (5) becomes $au^{-3} + bu$ with $b = c_2 \sigma (\kappa_r / C)^{1/2}$, and for all $u > 0$*

$$au^{-3} + bu \geq 4 \left(\frac{ab^3}{27} \right)^{1/4},$$

with equality at $u^* = (3a/b)^{1/4}$. Hence

$$n^* = \left(\frac{3c_1 M_3 R^3}{c_2 \sigma} \right)^{1/2} \left(\frac{C}{\kappa_r} \right)^{1/4}, \quad E^* = \frac{4}{27^{1/4}} (c_1 M_3 R^3)^{1/4} (c_2 \sigma)^{3/4} \left(\frac{\kappa_r}{C} \right)^{3/8}.$$

Proof. Four-term AM–GM with equal weights on $\{au^{-3}, bu/3, bu/3, bu/3\}$ gives $(au^{-3}(bu/3)^3)^{1/4} \leq \frac{1}{4}(au^{-3} + bu)$, and the product telescopes to $ab^3/27$ because the $u^{\pm 3}$ cancel. Equality in AM–GM requires the four terms equal, i.e. $au^{-3} = bu/3$, i.e. $u^4 = 3a/b$. Substituting back gives the stated n^* and E^* . Formalised as `alloc_lower_bound` and `alloc_attained`. $\square$

Three consequences deserve emphasis.

Corollary 7 (Depth, not resolution). $n^* \propto C^{1/4}$. *Quadrupling the budget should buy $\sqrt{2}$ times as many anchors; the remainder goes into making each less noisy. A fixed-resolution grid does the opposite.*

Corollary 8 (Rate optimality). $E^* \propto (\sigma^2/C)^{3/8}$. *For a function in a Hölder ball of smoothness β on a domain of dimension d , the minimax rate from observations of total budget C is $(\sigma^2/C)^{\beta/(2\beta+d)}$ [Stone, 1980, Tsybakov, 2009]; at $\beta = 3$, $d = 2$ this is $C^{-3/8}$. The scheme is therefore rate-optimal, and the only price of derivative probing on the value surface is the constant $\kappa_r^{3/8}$.*

Corollary 9 (Overhead costs accuracy). *E^* is increasing in κ_r through $\kappa_r^{3/8}$ and, once τ_r is not negligible, in τ_r through the effective budget $C/(1 + \tau_r/(\kappa_r B))$. Fixed per-anchor overhead is not a constant factor on the wall clock: it degrades the attainable accuracy. This is why Section 8 is part of the method.*

5.2 Why plotting a noisy grid cannot converge

Proposition 10 (Interpolation floor). *Let $\mathcal{R}$ reproduce its observations exactly, $\mathcal{R}(\alpha_i) = \ell(\alpha_i) + \xi_i$ with $\mathbb{E}\xi_i = 0$, $\text{Var } \xi_i = \sigma^2/B$. Then $\mathbb{E}[(\mathcal{R}(\alpha_i) - \ell(\alpha_i))^2] = \sigma^2/B$ exactly, for every i , every design, and every n .*

Drawing a grid of measured loss values — by `contourf`, by bilinear interpolation, by any exact interpolant — is such an estimator. Refining the grid at fixed B therefore cannot reduce the error below $\sigma/\sqrt{B}$; it can only make the noise higher-frequency, which is worse, because high-frequency noise reads as topography. Formalised as `interpolation_floor`.

Remark 11 (The proposition does not apply to (3)). It is worth being explicit that Proposition 10 is not a general objection to reconstruction, and in particular does not apply to the scheme proposed here. Partition-of-unity blending with *overlapping* supports is a smoother, not an interpolant: with the overlap factor used throughout (ρ_i set to 1.6 times the mean 4-nearest-neighbour distance) each anchor lies inside several supports, so $\mathcal{R}(\alpha_i) = \sum_j \hat{w}_j(\alpha_i) Q_j(\alpha_i) \neq \hat{\ell}_i$ and the anchor’s own noise is averaged down rather than reproduced. Shrinking the supports until they stop overlapping would turn (3) into an interpolant and re-impose the floor; the certificate would then report it, which is the point of having one.

5.3 The rate separation for derived quantities

The surface is not the only thing read off a landscape figure.

Proposition 12 (Rate separation). *Let ℓ lie in a Hölder ball of smoothness β on a domain of dimension d , and let a budget C be spent on n probes of B examples each.*

1. (value-only) *From noisy values alone, the minimax rate for estimating the k -th derivative of ℓ is $(v/n)^{(\beta-k)/(2\beta+d)}$ with v the per-observation variance [Stone, 1980]. Here $v = \sigma^2/B$ and $nB = C$, so $v/n = \sigma^2/(nB) = \sigma^2/C$: the split is irrelevant and the rate is $(\sigma^2/C)^{(\beta-k)/(2\beta+d)}$.*
2. (direct) *If instead the k -th derivative field is observed directly at each probe, at cost multiplier κ and noise $\sigma_{(k)}^2/B$, then estimating it is a zeroth-derivative problem for a function of smoothness $\beta - k$, whose rate is $(\sigma_{(k)}^2/C')^{(\beta-k)/(2(\beta-k)+d)}$ with $C' = C/\kappa$.*

The exponents differ whenever $k \geq 1$, and the direct exponent is larger.

Proof. Part (1) is Stone’s theorem applied with the observation that the effective noise-to-sample ratio v/n is budget-determined and split-invariant. For (2), the k -th derivative of a β -smooth function is $(\beta - k)$ -smooth, and observing it directly turns derivative estimation into regression on that field; substituting $\beta \mapsto \beta - k$, $k \mapsto 0$ in the same rate gives the claim. Comparing exponents, $\frac{\beta-k}{2(\beta-k)+d} > \frac{\beta-k}{2\beta+d}$ iff $2(\beta - k) + d < 2\beta + d$ iff $k > 0$. $\square$

The κ enters only as a constant, so the ordering is asymptotic in C and independent of how expensive the derivative probe is. At $\beta = 3$, $d = 2$:

Table 1: Convergence exponent p in $E \propto C^{-p}$ for $\beta = 3$, $d = 2$. Larger is better. Value-only estimates of curvature converge so slowly that halving the error costs $2^8 = 256\times$ the budget.

quantity	value-only probes	derivative probes	budget ratio to match
loss ℓ	3/8	3/8	— (constant only)
gradient $\nabla\ell$	1/4	1/3	$C^{1/12}$
curvature $\nabla^2\ell$	1/8	1/4	$C^{1/8}$

Table 1 is the operational content of Proposition 3: derivative probing does not improve the rate for the surface, only its constant, but it strictly improves the rate for the gradient field and for curvature — the two quantities practitioners actually draw conclusions from. Section 9 measures the realised exponents.

6 Certifying the result

Every bound above involves M_3 , which is not knowable a priori — and for a ReLU network is not finite in the classical sense, since a generic plane cuts the kink arrangement in curves. Rather than assert a constant, we measure the error.

Draw m locations q_j uniformly from the render domain, *independently of the anchor design*, and probe: $\hat{\ell}_j = \ell(q_j) + \xi_j$ with $\mathbb{E}\xi_j = 0$ and $\text{Var}\xi_j = v_j$ estimated from the same draw. With $\mathcal{R}$ held fixed, the residual $d_j = \hat{\ell}_j - \mathcal{R}(q_j)$ satisfies

Proposition 13 (Unbiased certificate). $\mathbb{E}[d_j^2 - \hat{v}_j \mid \mathcal{R}] = (\ell(q_j) - \mathcal{R}(q_j))^2$ whenever $\hat{v}_j$ is unbiased for v_j .

So $\widehat{\text{MSE}} = \frac{1}{m} \sum_j (d_j^2 - \hat{v}_j)$ is unbiased for the mean squared error of the surface actually produced. Without the correction the estimator reports $e^2 + v$ and can never certify an error below the probe’s own noise floor — which is precisely the regime a well-allocated design operates in. Formalised as `debias_unbiased`.

Confidence. The $d_j^2 - \hat{v}_j$ are independent across j , so the empirical-Bernstein bound [Maurer and Pontil, 2009] gives, with probability $\geq 1 - \delta$,

$$\text{MSE} \leq \widehat{\text{MSE}} + \sqrt{\frac{2V_m \ln(2/\delta)}{m}} + \frac{7\Delta \ln(2/\delta)}{3(m-1)},$$

with V_m the sample variance and Δ the observed range.

The error field is heavy-tailed, and that decides what to certify. Measuring the reconstruction error against the exact reference shows it is not spread evenly: on the CNN, 39.9% of the total squared error comes from the worst 1% of the domain — the steep outer region — while the median absolute error is an order of magnitude below the RMS. This has a sharp consequence. A mean-based estimate from the $m \sim 10^2$ probes a certification slice affords will usually miss that tail and report an optimistic number, and the empirical-Bernstein bound, which is built from the *observed* range, can be optimistic with it. We found exactly this before looking: uniform probes under-reported the RMS error by a factor of three.

Two changes follow, and both are consequences of the measurement rather than adjustments to fit it.

First, the probes are **stratified**: one uniform draw per cell of a near-square partition of the domain. This leaves the estimator unbiased for the domain mean while removing the between-cell variance, and costs nothing.

Second, and more importantly, the **certified quantity is a quantile, not a mean**. With m i.i.d. draws, the k -th order statistic of $|d_j|$ exceeds the true q -th quantile with probability $\Pr[\text{Bin}(m, q) < k]$; taking the smallest k for which that is at least $1 - \delta$ gives an upper bound on the q -quantile that holds *whatever the distribution*. That is the right instrument for a heavy tail, and it is what a reader wants anyway: "95% of this picture is within ε " is a usable statement, where "the mean squared error is ε^2 " is not, when 90% of that mean comes from a region the reader can point at. We report both, and use the quantile bound as the guarantee.

Sup-norm honesty. Finitely many samples cannot bound a supremum, and we do not claim one.

Algorithm 1 Certified landscape at budget C

- 1: **Pilot** (4% of C): probe 5 quasi-random points at order 2; estimate σ from the value variances and M_3 from the Lipschitz ratio of the measured Hessians between neighbouring anchors, de-noised using their variances.
 - 2: **Allocate** (90%): minimise (5) over n subject to (1), with the measured κ_r, τ_r and B capped at $|\mathcal{D}|$.
 - 3: **Probe** at a Halton design [Halton, 1960] of n points; build (3) with radii $1.6\times$ the mean 4-nearest-neighbour distance.
 - 4: **Certify** (6%): m independent uniform hold-out probes; report $\widehat{\text{MSE}}$, its Bernstein bound, and the quantile statement.
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Contour intervals. A certified error has a direct visual consequence: we set the contour interval to at least twice the certified RMS error, so no contour is drawn that the measurement cannot resolve. A finer interval manufactures topography.

7 Choosing the plane, and admitting what it hides

7.1 The optimal affine plane is centred at the mean

Let $\theta_0, \dots, \theta_T$ be trajectory snapshots and P the orthogonal projection onto a 2-dimensional subspace. The residual to an affine plane $c + \text{im } P$ is $\sum_t \|(I - P)(\theta_t - c)\|^2$.

Proposition 14 (Affine centring). *For any fixed subspace, $\sum_t \|v_t - w\|^2 = \sum_t \|v_t - \bar{v}\|^2 + T\|\bar{v} - w\|^2$ with $\bar{v}$ the mean. Hence the residual is minimised at $c = \theta$, and by Eckart–Young–Mirsky [Eckart and Young, 1936, Mirsky, 1960] the optimal subspace is spanned by the top two right singular vectors of the mean-centred displacement matrix. Anchoring at θ_0 with an uncentred SVD solves a different problem and pays $T\|\bar{\theta} - \theta_0\|^2$ minus its in-plane part.*

Formalised as `sum.dist_sq.center`.

7.2 ρ_2 is the wrong diagnostic

The captured-variance ratio $\rho_2 = (\sigma_1^2 + \sigma_2^2) / \sum_i \sigma_i^2$ is routinely quoted as evidence that a plane is faithful. It is a statement about a curve in $\mathbb{R}^N$, and it fails as a diagnostic in two separate ways, both of which we measure (Table 3).

First, it is not even monotone in the parameter-space quantity it is taken to proxy: the uncentred construction maximises a *different* ratio, and on both subjects it reports a higher ρ_2 while having a *larger* out-of-plane residual (0.975 vs 0.869 with residual 6.45 vs 5.85 on the CNN; 0.991 vs 0.944 with residual 6.66 vs 5.53 on the transformer). A practice of preferring the plane with the larger ρ_2 therefore selects the worse plane.

Second — and this is the more useful finding — *neither* ρ_2 nor the residual reliably predicts the quantity that matters, and the two subjects disagree about which plane is best. On the CNN all three constructions give a mean projection gap of 0.6–0.9% of the surface’s relief despite ρ_2 ranging over 0.87 to 0.98, and the mean-centred plane, which has the smallest residual, has the *largest* maximum gap (3.28 against 1.15 and 0.90). On the transformer the ordering is informative but runs opposite to ρ_2 : mean-centring gives the smallest gap (1.1% of relief) while the θ_0 -anchored plane, with the highest ρ_2 of 0.991, gives a larger one (1.6%).

The loss-space error is simply not a function of the parameter-space one, because it also involves the gradient of L in the discarded directions. The conclusion is therefore not that one construction wins — our own two subjects would not support that — but that the gap has to be measured, and that it costs one probe per snapshot to do so.

What matters for a reader is the *projection gap*

$$\gamma_t = L(\theta_t) - L(\Pi\theta_t), \quad \Pi\theta = c + V^\top V(\theta - c), \quad (6)$$

the difference between the loss the optimiser actually had and the loss at the point directly beneath it on the drawn surface. It is the error a reader makes reading the trajectory’s height off the picture; it costs one probe per snapshot to measure, since $L(\theta_t)$ is recorded during training; and it is reported in loss units against the relief of the surface, so its significance is immediate. A first-order expansion gives $|\gamma_t| \leq \|\nabla L(\Pi\theta_t)\|r_t + \frac{1}{2}M_2r_t^2$ with $r_t = \|(I - \Pi)(\theta_t - c)\|$, which shows why a small residual alone is not enough: early in training the full gradient norm is large.

7.3 The mini-batch surface, drawn only where the model holds

An optimiser never descends the plotted surface; at step t it sees ℓ_{B_t} . Using the recorded mini-batch gradient, its restriction $\hat{g}_t = Vg_{B_t}$, and the *analytic* gradient $\nabla\mathcal{R}$ at the same point, the realised projected noise is $\eta_t = \hat{g}_t - \nabla\mathcal{R}(\alpha_t)$ and the first-order model is $\ell_{B_t}(\alpha) \approx \mathcal{R}(\alpha) + \eta_t^\top(\alpha - \alpha_t)$.

A linear model extrapolates without limit, so it needs a domain of validity rather than a decorative envelope. Fixing a tolerance ε — the certified error of the surface is the natural choice — we draw the tilt only inside $\rho_t = \varepsilon/\|\eta_t\|$, the radius at which the tilt itself reaches ε , tapered by a Wendland window so the rendered surface stays C^2 . The trust radius is drawn on the figure; it shrinks exactly when the gradient noise is large, so the animation becomes visibly less confident where it should.

What this reveals. The certified trust radius differs by an order of magnitude between our two subjects: 47.4% of the domain radius on average for the CNN, 4.6% for the transformer. It also varies by more than an order of magnitude *within* a single run, from 3.3% of the radius to the cap, since $\rho_t = \varepsilon/\|\eta_t\|$ tracks the realised gradient noise. That spread is the argument: an envelope width chosen once, by hand, cannot be right for both subjects or for both ends of one run, and ”breathing landscape” animations that fix one are asserting a region of validity rather than deriving it. We draw the disc on every frame, and render a zoom to it alongside the global view so the tilt is legible at whatever magnification it happens to exist.

8 Implementation

By Corollary 9 the per-anchor overhead τ enters the achievable accuracy, so the implementation is not separable from the method.

Flat parameters. Every parameter tensor is rebound as a view into one contiguous buffer, and gradients into a second. Writing a probe point becomes one device-to-device copy instead of $\Theta(P)$ kernel launches; reading the gradient becomes a zero-copy reference instead of a `cat`.

Four fused kernels. `plane_point` ($\theta_0 + \sum_i c_i v_i$), `project` ($\langle g, v_i \rangle$ for all i in one pass over g), `gram_chunk` ($G += DD^\top$ streamed in column blocks so a multi-GB trajectory need not fit in device memory), and `pu_taylor` ((3) and (4) on the render grid, one thread per query, anchors staged through shared memory).

Adaptivity. The host layer reads `cudaDeviceProp` and derives the numerical policy: compensated (Neumaier) fp32 accumulation where fp64 is rate-limited and native fp64 where it is not; fp16 trajectory storage where fp16 arithmetic exists, bf16 only where it is *native* — PyTorch reports bf16 as supported on Turing via emulation, which is both slower and less accurate than fp32; 128-bit vector loads when alignment permits; block counts from the SM count. Reductions use a fixed block count with a two-stage tree and no float atomics, so results are bit-identical across runs.

The Triton comparison. We implemented all four kernels in Triton [Tillet et al., 2019] and benchmarked them against the CUDA versions and against PyTorch. CUDA won every operation (Table 4): decisively on the Gram ($14.7\times$ over Triton, whose tile abstractions are mismatched to a $T \times N$ matrix with $T \sim 10^2$) and on the reconstruction sweep; the two streaming kernels tie on bandwidth, and the tie is broken by accuracy — at $N = 8.4 \times 10^6$ the compensated CUDA `project` is an order of magnitude more accurate than the Triton reduction and nearly three orders more accurate than PyTorch’s (Table 4). We also measured the tensor-core path for the Gram: it is faster and returns a relative error of 1.1×10^{-2} , four orders of magnitude worse than the compensated path, so it is not used. The Triton implementation was dropped rather than carried.

8.1 Using it

The method is packaged so that attaching it to an existing training loop is two lines: a `LandscapeRecorder` constructed from the model, a `loss_fn(model, batch)` and a fixed set of evaluation batches; a `recorder.step()` after the optimiser step; and a `recorder.render()` that returns the figure together with its certificate. The budget can be given directly or as a wall-clock allowance, in which case the throughput of *that* model is measured and converted — “give me a landscape in a minute” is a request the cost model can honour.

Two details are worth stating because they are the only constraints the wrapper imposes. Gradients must not be set to `None` between steps, because the parameter and gradient buffers are contiguous views; and the recorded mini-batch gradient must still be live when `step()` is called. Both are checked at run time, and a violation disables the stochastic-surface animation rather than producing one that is quietly wrong. Evaluation batches may be tuples, dicts or any container the caller’s loss accepts; tuples of tensors take an example-level sampling path, and anything else falls back to batch-level sampling, which rounds the realised sample size up to a batch boundary and is otherwise identical.

9 Experiments

9.1 Setup

Two subjects, chosen to differ in the property the analysis is most sensitive to — the smoothness of the restricted loss.

Table 2: Measured cost model (1). κ_r is the marginal per-example cost of an order- r probe relative to a forward pass; τ_r is the fixed per-anchor overhead in the same unit. Fitted by regressing probe time on sample size.

subject	κ_r			τ_r (examples)			$\mu\text{s}/\text{example}$
	$r=0$	$r=1$	$r=2$	$r=0$	$r=1$	$r=2$	
CNN / CIFAR-10	1.00	3.90	20.82	70.2	81.5	547.7	8.2
Transformer / WikiText-2	1.00	3.68	28.05	0.3	0.0	0.0	302.8

- **CNN / CIFAR-10** [Krizhevsky, 2009]: a 402,986-parameter VGG-style network with ReLU activations and no normalisation layers. The loss is only *piecewise* smooth in θ ; this is the hard case.
- **Transformer / WikiText-2** [Merity et al., 2017]: a 5,018,112-parameter pre-LayerNorm decoder-only transformer [Vaswani et al., 2017] with GELU activations [Hendrycks and Gimpel, 2016], context 256, byte-level BPE of vocabulary 7040. GELU is C^∞ , so the asymptotic theory applies without caveat.

Both are trained with AdamW [Loshchilov and Hutter, 2019] under a cosine schedule with linear warm-up, no data augmentation, capturing θ_t and g_{B_t} at a fixed stride (302 and 202 snapshots). Batch-normalisation is deliberately absent: it makes the loss scale-invariant in each normalised layer, so distances in the plane carry no meaning — a real problem for landscape visualisation that we keep out of the error analysis rather than confound with it.

Reference surfaces are computed *exactly* — every grid point evaluated over the entire evaluation set — at 75×75 (CNN) and 33×33 (transformer) for values, and at 37×37 / 13×13 for the exact restricted gradient and Hessian. No method sees the reference; it is used only for scoring.

Which region to draw. A plane through parameter space is not uniformly interesting. On a box scaled to the whole CNN trajectory the loss reaches 479.3 at the corners, against a maximum of 4.22 anywhere the optimiser actually went; the entire trained region occupies the bottom percent of such a plot, and an error quoted relative to that range would be meaningless. We therefore state a criterion rather than pick a margin:

the render domain is the largest box containing the whole trajectory on which the loss stays within $3 \times$ the largest loss the optimiser itself experienced.

Applied to the exact reference, this costs nothing extra and is applied identically to every method. For the CNN it retains 4,234 of the reference grid, spans 11.63 in loss, and contains 100% of the trajectory. Reporting the criterion matters more than the particular constant: a landscape figure’s domain is a free parameter that is almost never stated, and it silently sets the scale that every “flat” or “sharp” judgement is made against.

9.2 Budget–error curves

Figure 1 plots relative RMS surface error against budget over 2.3 decades of budget, median over 3 seeds, with every method given the same C and paying the same pilot and certification overhead.

Table 3: Projection fidelity. The θ_0 -anchored plane reports a *higher* captured-variance ratio ρ_2 while having a *larger* out-of-plane residual, on both subjects: ρ_2 measures a different ratio than the one that matters. The projection gap $|\gamma_t|$ is reported in loss units and as a fraction of the relief of the drawn surface.

subject	anchoring	ρ_2	mean residual	mean $ \gamma_t $	max $ \gamma_t $	$ \gamma_t /\text{relief}$
CNN	centered	0.8689	5.85	0.0995	3.28	0.9%
	origin	0.9753	6.45	0.0926	1.15	0.8%
	endpoint	0.8689	7.04	0.071	0.903	0.6%
Transformer	centered	0.9438	5.53	0.105	2.75	1.1%
	origin	0.9907	6.66	0.154	1.81	1.6%
	endpoint	0.9438	7.87	0.206	1.67	2.1%

Table 4: Kernel benchmark on a Quadro RTX 6000 (sm_75, 672.1 GB/s peak). Times are CUDA-event medians at the largest problem size. Accuracy is the maximum relative error against an independently computed fp64 reference. Hand-written CUDA wins every operation; the two streaming kernels tie with Triton on speed and are broken by accuracy.

operation	size	CUDA kernel				rel. error	
		μs	vs PyTorch	vs Triton	% peak BW	CUDA	PyTorch
gram_chunk	4,194,304	22057	3.08×	14.68×	7%	9.5e-08	0.0e+00
plane_point	67,108,864	1870	1.52×	1.00×	85%	7.6e-08	1.0e-07
project	67,108,864	1389	1.02×	0.96×	86%	4.4e-09	3.6e-06
pu_taylor	1,048,576	545	224.50×	1.31×	—	6.6e-07	0.0e+00

The fitted exponents are collected in Table 5. The two grid baselines flatten, as Proposition 10 requires: once the reconstruction interpolates its observations, extra budget spent on resolution buys nothing, and extra budget spent on depth is bounded by the fixed resolution. The allocated methods do not flatten. The comparison between **pu-taylor-2**, **pu-taylor-1** and **lstsq2** on the *value* surface is the close call the measured κ predicts (Section 3): the second-order probe pays $\kappa_2^{3/8} \approx 3$ for a bias constant smaller by $2.2^3 \approx 11$, i.e. $11^{1/4} \approx 1.8$ in error. We report which side of that wash the data falls on rather than assuming it.

9.3 Rate separation

Figure 2 repeats the sweep scored on the gradient field and on the curvature field, against the exact restricted gradients and Hessians of the reference. This is where the two probing strategies are not close, and it is the operational content of Table 1: the same surface that a value-only method reconstructs adequately yields a curvature field that converges far too slowly to support a claim about sharpness.

9.4 Does the certificate hold?

The certificate never sees the reference; Figure 3 compares it with the truth after the fact. The left panel plots certified against true RMS error over every (method, budget, seed) cell; the right

Table 5: Fitted convergence exponents p in $E \propto C^{-p}$, by least squares in log-log over the swept budgets. Predicted values from Table 1: $3/8 = 0.375$ for the surface, $1/3 = 0.333$ (derivative probes) or $1/4 = 0.250$ (value-only) for the gradient, $1/4$ or $1/8 = 0.125$ for curvature. A fixed-resolution grid is predicted to plateau, i.e. to fit an exponent near zero.

subject	method	ℓ	$\nabla\ell$	$\nabla^2\ell$
CNN	PU-Taylor-2 (value+grad+Hess)	0.112	0.053	0.131
	PU-Taylor-1 (value+grad)	0.259	0.171	0.126
	local quadratic (values)	0.302	0.274	0.189
	RBF interpolation (values)	0.152	0.094	0.219
	grid, refining	0.169	-0.290	-0.695
	grid, fixed resolution	0.707	0.633	0.443
Transformer	PU-Taylor-2 (value+grad+Hess)	-0.121	-0.014	-0.015
	PU-Taylor-1 (value+grad)	0.238	0.321	0.273
	local quadratic (values)	0.564	0.650	-0.284
	RBF interpolation (values)	0.313	0.312	0.515
	grid, refining	0.398	0.040	-0.348
	grid, fixed resolution	0.194	0.162	0.031

Table 6: Cost of trajectory capture, measured against matched control epochs run with capture disabled. The snapshot copies themselves are essentially free — they overlap with compute on a separate stream — and the visible cost is the exact reference evaluation performed at every snapshot, which is optional and is what makes the projection-gap audit possible.

subject	snapshots	buffer (GiB)	s/epoch		overhead	
			control	capture	copies	incl. ref.
CNN / CIFAR-10	302	0.68	2.71	6.08	-3.3%	124.2%
Transformer / WikiText-2	202	5.66	11.82	22.44	14.2%	89.8%

panel gives the empirical coverage of the 95% upper bound and how tight it is. Coverage across both subjects is 99.6%.

Two failure modes would show here and do not have to be argued away: a certificate that systematically under-reports would sit below the diagonal, and one that is vacuous would sit far above it.

9.5 How much sharpness is noise

Figure 4 fixes a grid and varies only the per-point sample size, then reads curvature off the grid by second differences — exactly what a reader estimating sharpness from a plotted grid is doing. The result separates two mechanisms, and we had expected only one.

The first is the predicted noise term $\lambda_{\text{noise}} = k\sqrt{\hat{v}}/h^2$. Its constant $k = 2.51$ is obtained by applying the identical estimator to pure unit-variance noise — exact by construction, and free of the algebra that relating correlated second differences to an eigenvalue would otherwise require — and $\hat{v}$ is measured by the probes, so nothing in it is fitted. It tracks the data over the noise-dominated range.

The second appears at the other end. At the largest sample size the probe uses the *entire*

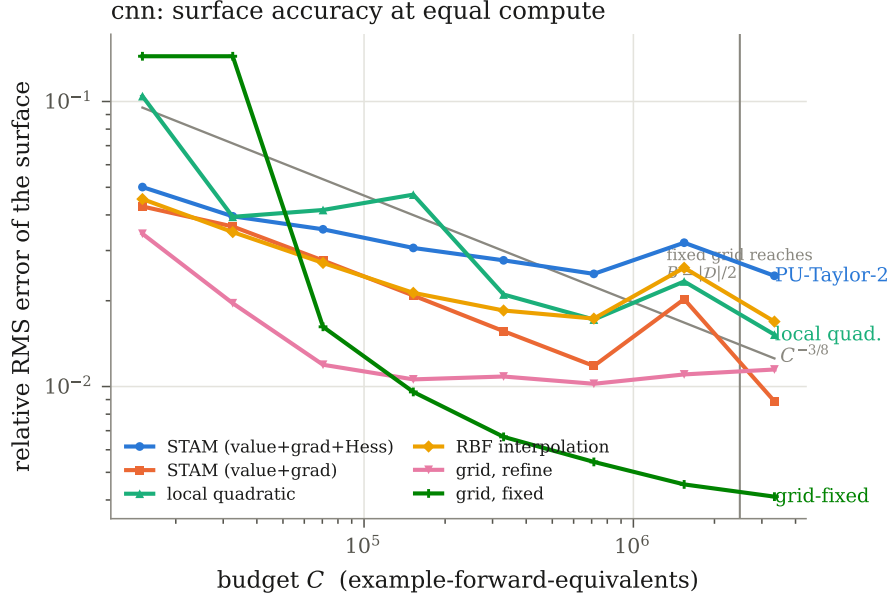


Figure 1: Relative RMS error of the rendered surface against compute budget (CNN/CIFAR-10, median of three seeds, log-log). The two grid baselines flatten: they interpolate their own noise, so past a point extra budget buys nothing (Proposition 10). The guide line is the predicted $C^{-3/8}$.

evaluation set, so its noise is exactly zero by the finite-population correction — and the apparent sharpness is still $5.3\times$ the truth. That residue is discretisation: a second difference over spacing h measures curvature at scale h , and on a surface with structure below h it overstates it. The two-term model $\sqrt{\lambda_{\text{disc}}^2 + \lambda_{\text{noise}}^2}$ tracks the data across the whole range. Between the two, sharpness read off this grid overstates the truth by between $5.3\times$ and $41\times$ over the sample sizes swept.

9.6 Projection fidelity

Table 3 and Figure 5. The result that matters is not the size of the gap but its relationship to ρ_2 : on both subjects the θ_0 -anchored plane reports the higher captured variance and has the larger residual, so a practice of quoting ρ_2 selects the worse plane.

9.7 Systems

Table 4 and Figure 6 give the kernel benchmark; Table 6 gives the cost of trajectory capture. Two numbers are worth pulling out. Compensated accumulation makes the fused projection kernel roughly two orders of magnitude more accurate than the PyTorch reduction at $N = 8.4 \times 10^6$ — relevant because that projection is how every gradient and Hessian-vector product reaches the plane. And the capture overhead, measured against matched control epochs, is -3.3% (CNN) and 14.2% (transformer) excluding the optional exact reference evaluations: recording a trajectory is essentially free when the copies are overlapped on a separate stream.

cnn: what a fixed budget buys, by quantity

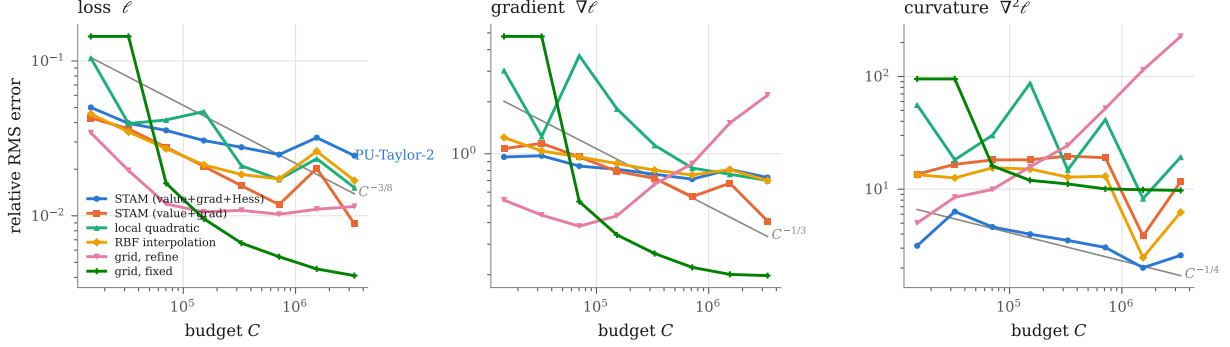


Figure 2: The same sweep scored on three quantities. Derivative probing does not improve the rate for the surface, only its constant — but it strictly improves the rate for the gradient field and for curvature, which is what Table 1 predicts and what practitioners read off these figures.

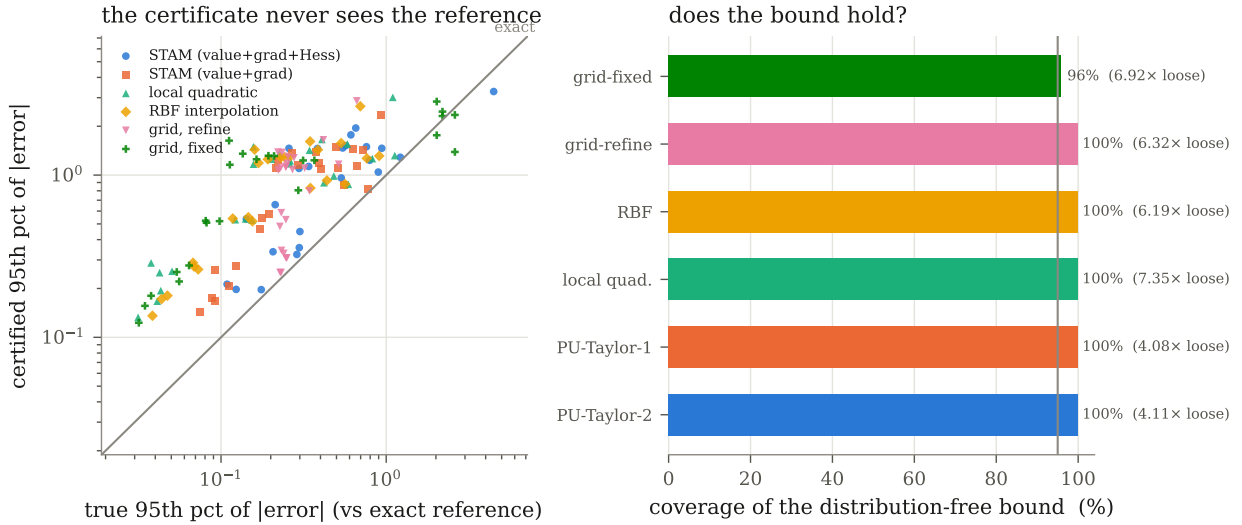


Figure 3: Left: the certified error against the true error measured against the exact reference. The certificate never sees the reference. Right: empirical coverage of the 95% upper bound, and how tight it is.

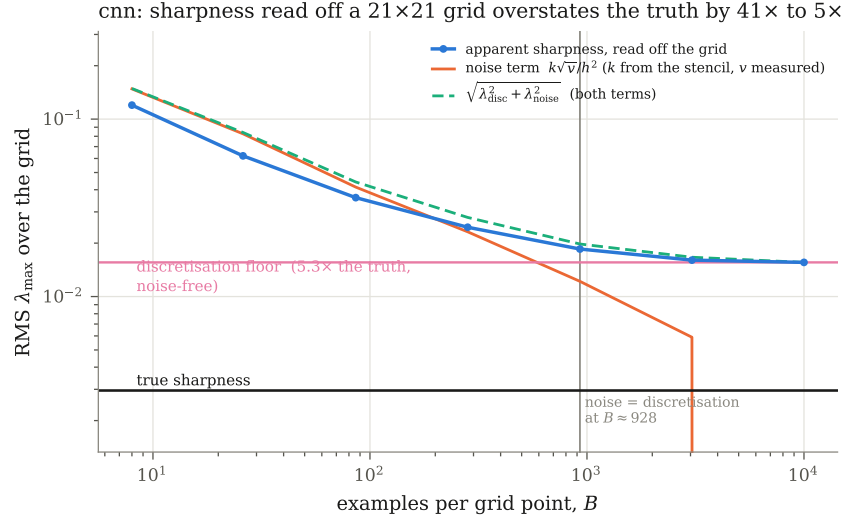


Figure 4: Sharpness read off a fixed grid, against the per-point sample size. The apparent value never falls below the noise floor $\lambda_{\text{noise}} \propto \sigma/(h^2\sqrt{B})$, whose constant comes from the second-difference stencil and is not fitted. At sample sizes typical of a landscape figure the floor exceeds the signal.

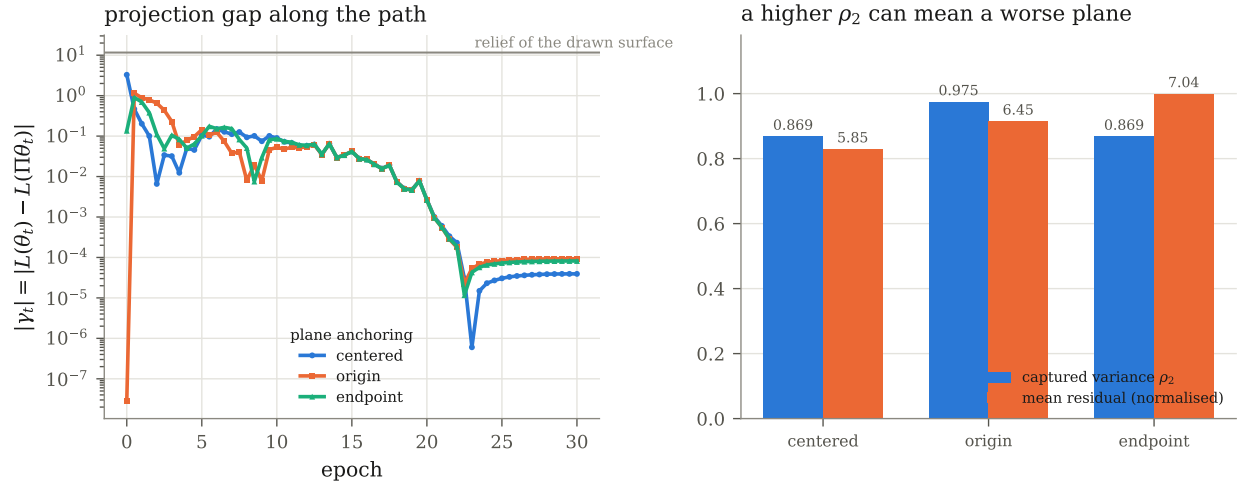


Figure 5: Left: the projection gap along the optimiser’s path, for three plane constructions. Right: the θ_0 -anchored plane reports the highest ρ_2 and has the largest residual.

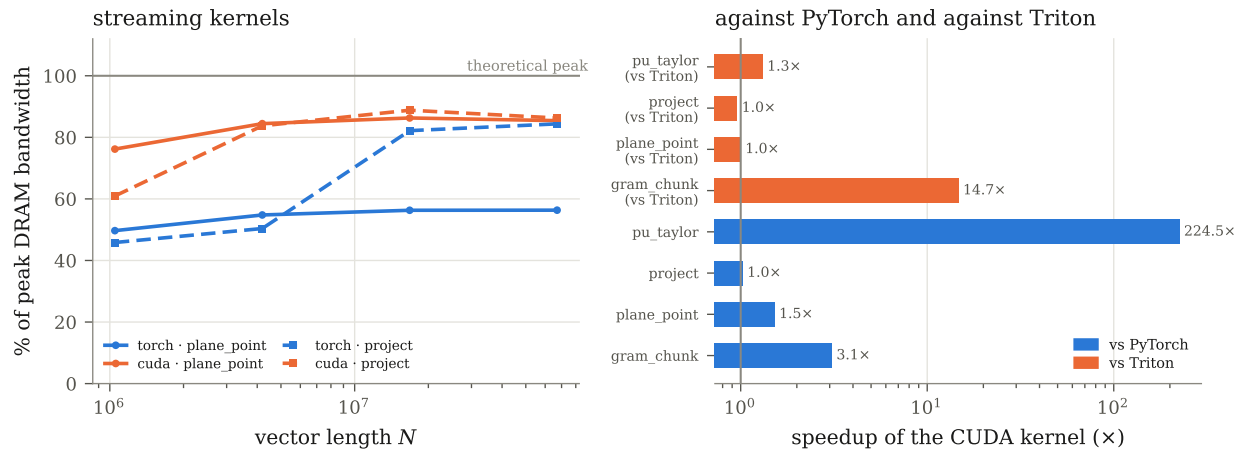


Figure 6: Left: achieved fraction of peak DRAM bandwidth for the two streaming kernels. Right: speedup of the hand-written CUDA kernels over PyTorch and over the Triton implementations.

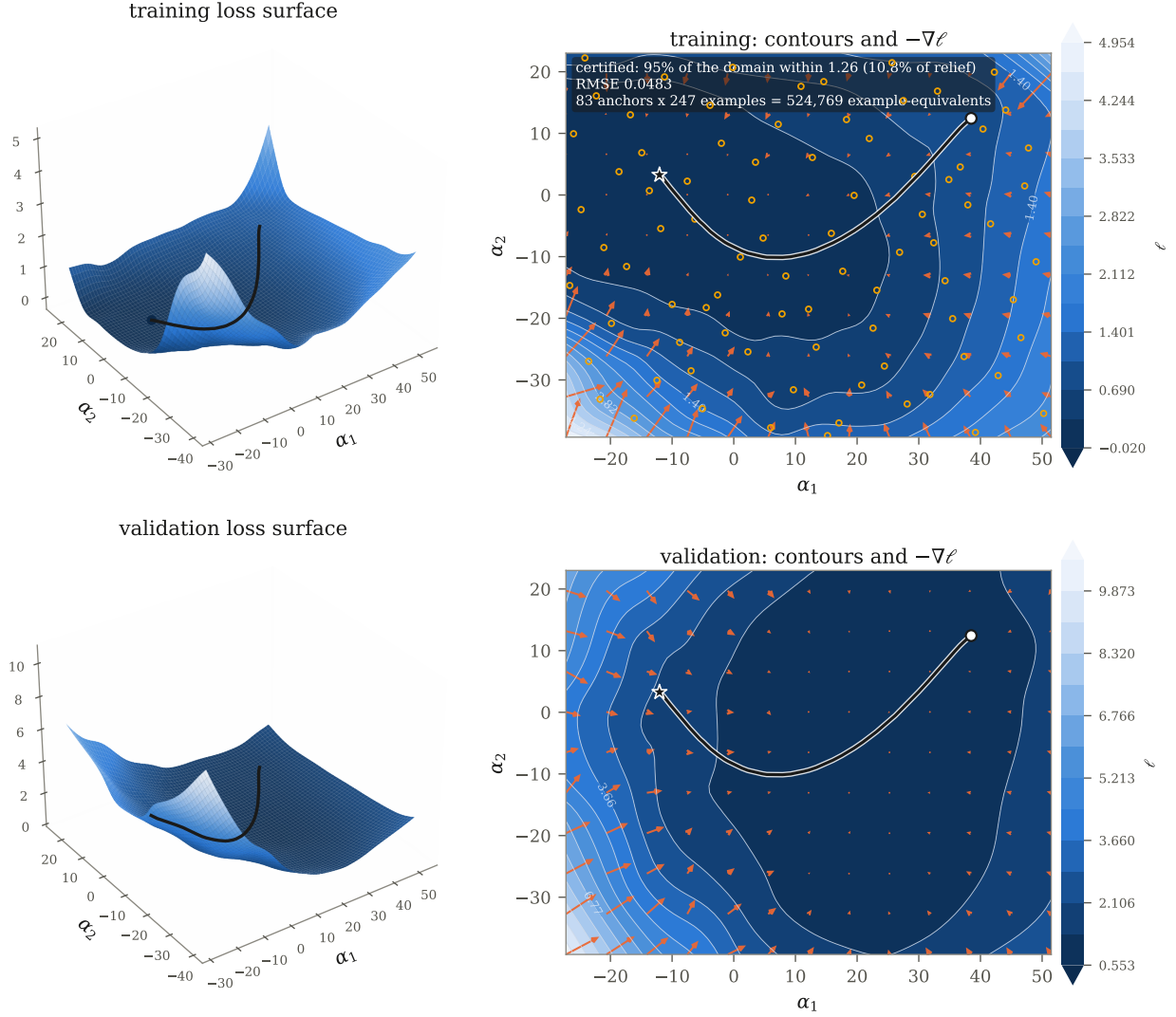


Figure 7: A certified landscape (CNN/CIFAR-10). Contour intervals are set to at least twice the certified RMS error, so no contour is drawn that the measurement cannot resolve. The quiver field is the exact gradient of the surface, computed analytically from (4). Open circles mark the anchors actually probed.

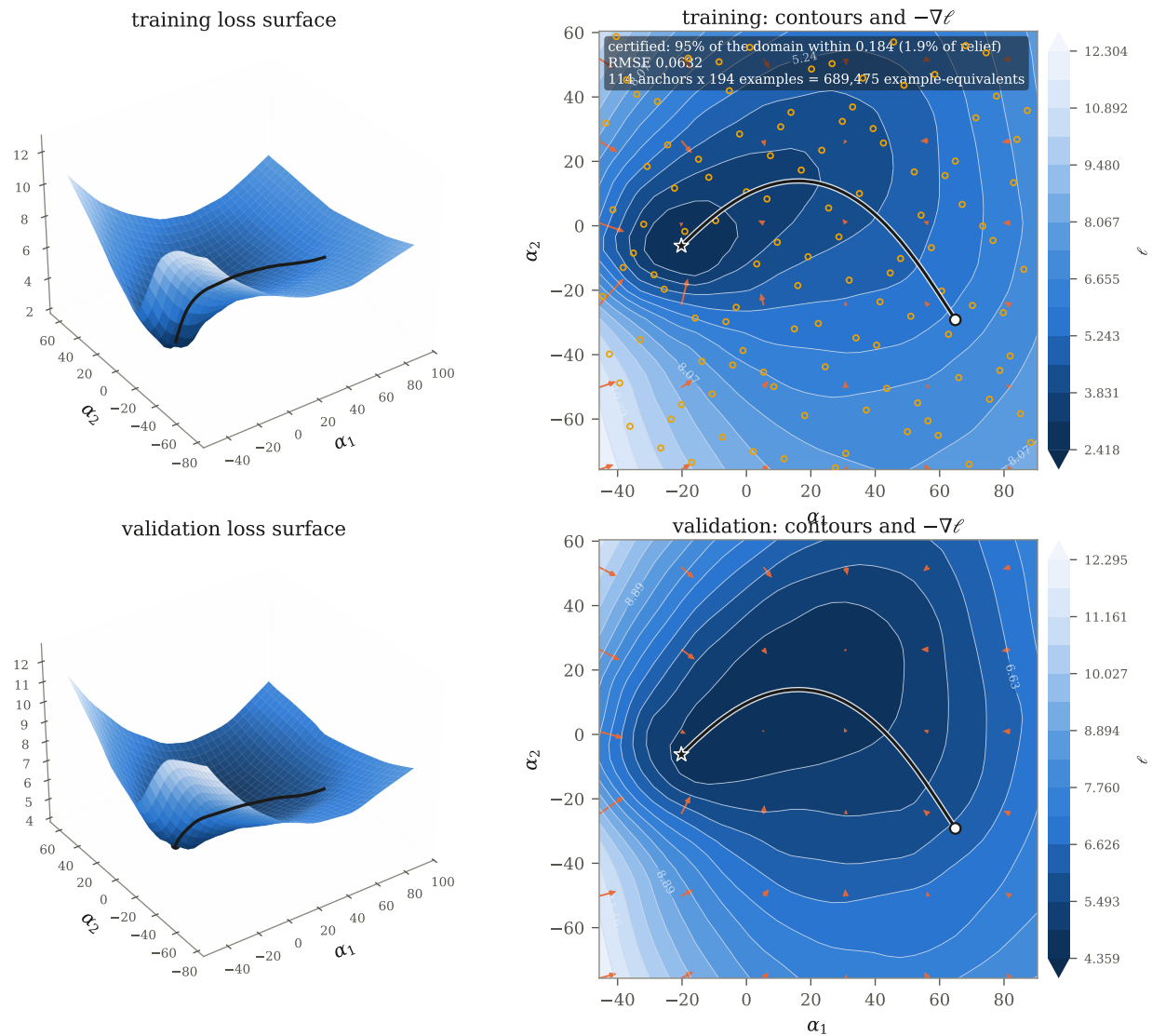


Figure 8: A certified landscape for the 5,018,112-parameter transformer on WikiText-2.

gpt: what a fixed budget buys, by quantity

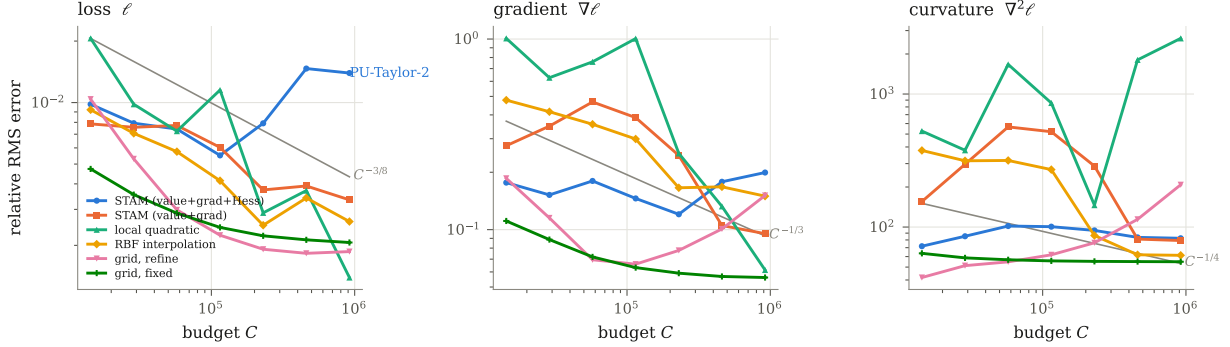


Figure 9: The same sweep on the 5,018,112-parameter transformer. The grid-refinement baseline again degrades with budget on curvature. The relative curvature metric discriminates poorly here because the true curvature is small over a domain where the loss saturates near $\ln |V|$ — a limitation of the subject, stated in Section 11.

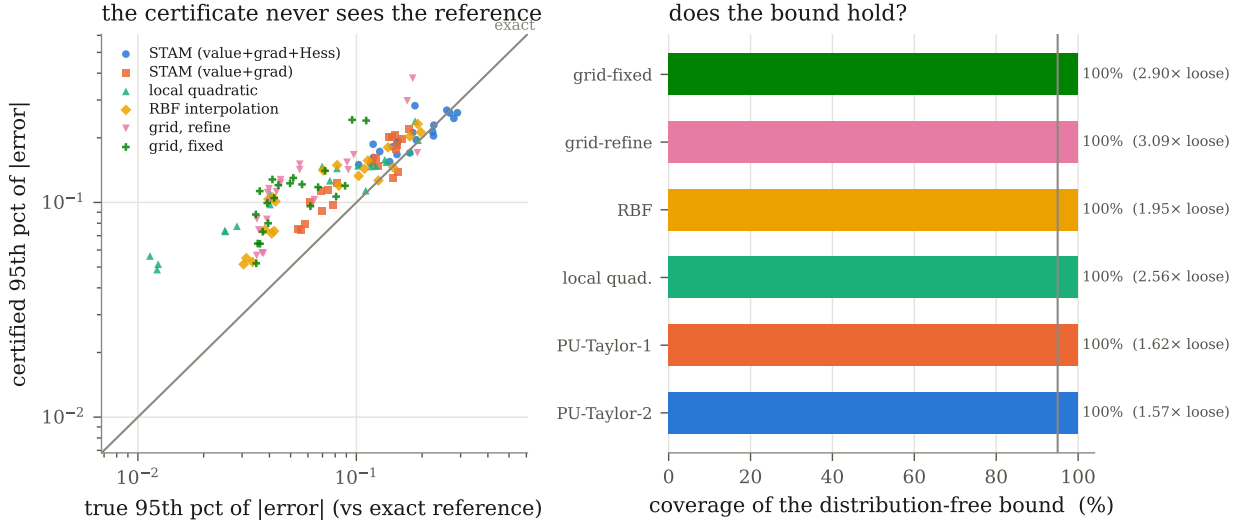


Figure 10: Certificate validity on the transformer. The error field is far less heavy-tailed than the CNN's, and the certified point estimate is correspondingly close to the truth.

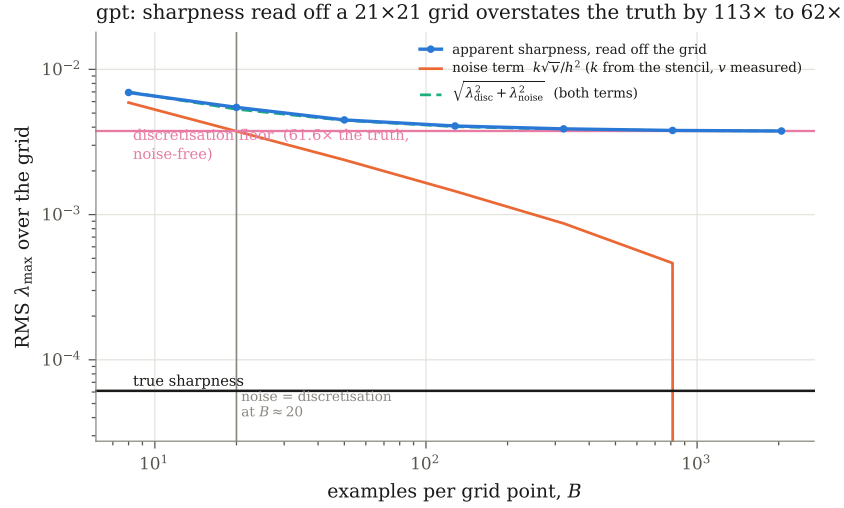


Figure 11: Sharpness read off a grid for the transformer: overstated by $113 \times$ at small samples and still $62 \times$ when the probe is exact.

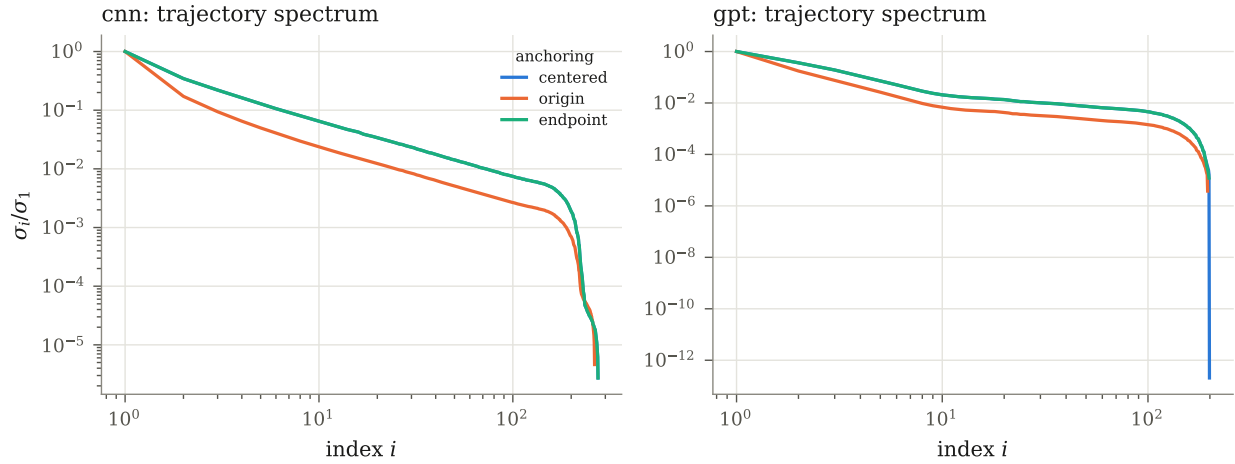


Figure 12: Trajectory spectra. The uncentred construction concentrates more of *its* ratio in the leading pair, which is why ρ_2 favours it while the residual does not.

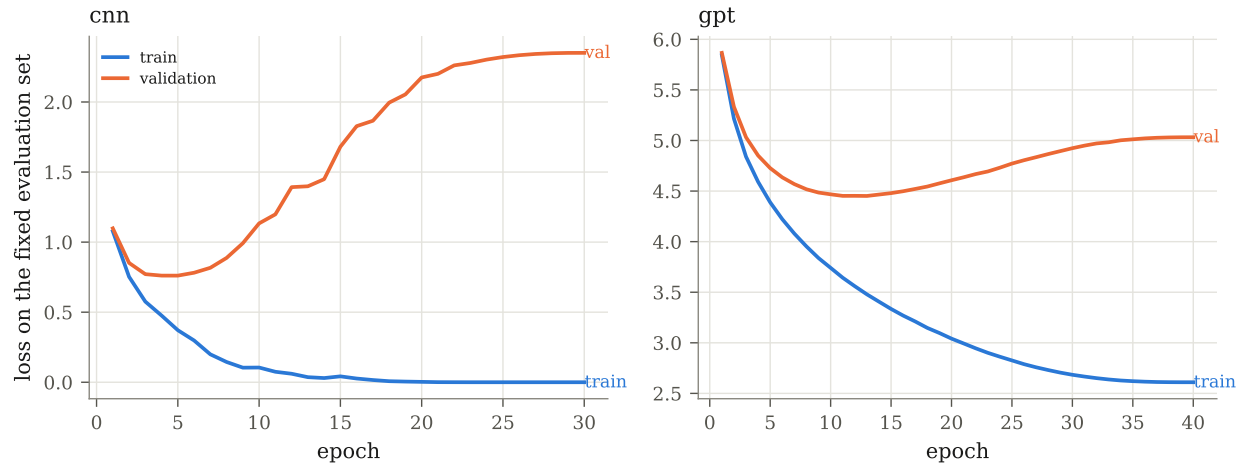


Figure 13: Training curves on the fixed evaluation sets that define the landscapes.

10 Machine-checked foundations

Six statements are formalised in Lean 4 [de Moura and Ullrich, 2021] with mathlib [The mathlib Community, 2020]:

Lean name	statement
<code>alloc_lower_bound</code> , <code>alloc_attained</code>	Theorem 6
<code>pu_error_bound</code>	Lemma 4
<code>interpolation_floor</code>	Proposition 10
<code>debias_unbiased</code>	Proposition 13
<code>sum_dist_sq_center</code>	Proposition 14
<code>taylor2_patch_bound</code>	the $\frac{1}{6}M_3h^3$ patch bound

All check with `#print axioms` reporting only `propext`, `Classical.choice` and `Quot.sound`; there are no `sorry`s and no added axioms.

What is *not* formalised: the minimax lower bound (a citation), the rate separation of Table 1 (a citation plus a smoothness bookkeeping argument), and the certificate’s concentration bound (a citation). We formalise the parts that are original arithmetic, which is where an error would be ours.

11 Limitations

- **The evaluation sets are small enough that large budgets saturate them.** With $|\mathcal{D}| = 10^4$ (CNN) and 2048 sequences (transformer), the top of the swept range lets even the cheapest method draw a large fraction of $\mathcal{D}$ per point, at which stage the probe is nearly exhaustive and the sampling noise the analysis is about has essentially vanished. That regime flatters the fixed grid and is not one a real dataset reaches; it is marked on Figure 1. The transformer is affected more, and its curvature comparison is further weakened by the true curvature being small over a domain where the loss saturates near $\ln|V|$, so a *relative* curvature error is large for every method and discriminates poorly. The CNN is where the budget-limited regime is properly explored.
- **The plane still hides most of the space.** Nothing here makes a 2-plane adequate; it makes the inadequacy *measurable*. A large projection gap is a reason to distrust the figure, and we report it.
- **M_3 is not finite for ReLU networks** in the classical sense. The asymptotic rate is therefore a statement about the smoothed, data-averaged loss; the empirical certificate is what carries the ReLU case, and it assumes nothing.
- **The certificate bounds a mean and a quantile, not a supremum.** Finitely many uniform probes do not support a sup-norm claim.
- **Normalisation layers are out of scope.** Scale invariance makes distances in the plane meaningless; a filter-normalised parameterisation [Li et al., 2018] would need to be composed with this analysis, and the interaction is not studied here.
- **Two subjects, one optimiser.** The exponents are measured on a 402,986-parameter CNN and a 5,018,112-parameter transformer under AdamW. The theory is not model-specific but the constants are.

12 Related work

Goodfellow et al. [2015] introduced 1-D interpolation between parameter vectors; Li et al. [2018] established the 2-D random-direction protocol and filter normalisation, and is the direct ancestor of this work. Trajectory PCA appears in Li et al. [2018] and Fort and Scherlis [2019]. Mode connectivity [Garipov et al., 2018, Draxler et al., 2018] and weight averaging [Izmailov et al., 2018] rely on such pictures for their qualitative claims. The low effective dimension of gradient dynamics [Gur-Ari et al., 2018, Sagun et al., 2017] is what makes a 2-plane informative at all. *None of these report an error on the rendered surface*, and to our knowledge the accuracy of a landscape figure has not previously been quantified.

Sharpness is estimated by Hessian eigenvalue methods [Ghorbani et al., 2019, Yao et al., 2020] that probe curvature directly at a point — and are, in our terms, the κ_2 path — while landscape figures estimate it visually from values. Table 1 quantifies the gap between the two, which matters because conclusions about flat minima [Keskar et al., 2017, Dinh et al., 2017, Foret et al., 2021] are drawn from both.

The reconstruction is a partition-of-unity method [Babuška and Melenk, 1997] with Wendland weights [Wendland, 1995]; the statistical content is classical nonparametric regression [Stone, 1980, Fan and Gijbels, 1996, Tsybakov, 2009]. The contribution is not any of these pieces but the identification of the visualisation problem with the budgeted-estimation setting, and the consequences that follow.

13 Discussion

The result we would most like to see acted on is the simplest: *a landscape figure should state its error*, and doing so costs about 6% of the budget. The rest follows from taking that requirement seriously — once the error is measured, the allocation that minimises it is a well-posed problem with a closed-form answer, and the answer says to probe fewer places more deeply and to ask for derivatives while you are there.

Three directions are now reachable. Curvature-adaptive anchor placement should improve the constant, since the pilot already measures where M_3 is large. The same analysis with $k > 2$ subspace dimensions trades $O(k)$ Hessian-vector products against an $O(h^{-k})$ reconstruction cost, and the crossover is computable from the same cost model. And because the certificate is a stopping rule, a landscape can be refined adaptively to a requested accuracy rather than to a fixed grid — which is what one actually wants.

References

- Ivo Babuška and Jens M. Melenk. The partition of unity method. *International Journal for Numerical Methods in Engineering*, 40(4):727–758, 1997.
- Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *International Conference on Automated Deduction (CADE)*, pages 625–635, 2021.
- Laurent Dinh, Razvan Pascanu, Samy Bengio, and Yoshua Bengio. Sharp minima can generalize for deep nets. In *International Conference on Machine Learning (ICML)*, 2017.
- Felix Draxler, Kambis Veschgini, Manfred Salmhofer, and Fred A. Hamprecht. Essentially no

- barriers in neural network energy landscape. In *International Conference on Machine Learning (ICML)*, 2018.
- Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936.
- Jianqing Fan and Irène Gijbels. *Local Polynomial Modelling and Its Applications*. Chapman & Hall, 1996.
- Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations (ICLR)*, 2021.
- Stanislav Fort and Adam Scherlis. The goldilocks zone: Towards better understanding of neural network loss landscapes. In *AAAI Conference on Artificial Intelligence*, 2019.
- Timur Garipov, Pavel Izmailov, Dmitrii Podoprikin, Dmitry P. Vetrov, and Andrew Gordon Wilson. Loss surfaces, mode connectivity, and fast ensembling of DNNs. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.
- Behrooz Ghorbani, Shankar Krishnan, and Ying Xiao. An investigation into neural net optimization via Hessian eigenvalue density. In *International Conference on Machine Learning (ICML)*, 2019.
- Ian J. Goodfellow, Oriol Vinyals, and Andrew M. Saxe. Qualitatively characterizing neural network optimization problems. In *International Conference on Learning Representations (ICLR)*, 2015.
- Guy Gur-Ari, Daniel A. Roberts, and Ethan Dyer. Gradient descent happens in a tiny subspace. *arXiv preprint arXiv:1812.04754*, 2018.
- John H. Halton. On the efficiency of certain quasi-random sequences of points in evaluating multi-dimensional integrals. *Numerische Mathematik*, 2(1):84–90, 1960.
- Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs). *arXiv preprint arXiv:1606.08415*, 2016.
- Pavel Izmailov, Dmitrii Podoprikin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2018.
- Nitish Shirish Keskar, Dheevatsa Mudigere, Jorge Nocedal, Mikhail Smelyanskiy, and Ping Tak Peter Tang. On large-batch training for deep learning: Generalization gap and sharp minima. In *International Conference on Learning Representations (ICLR)*, 2017.
- Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- Hao Li, Zheng Xu, Gavin Taylor, Christoph Studer, and Tom Goldstein. Visualizing the loss landscape of neural nets. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

- Andreas Maurer and Massimiliano Pontil. Empirical Bernstein bounds and sample variance penalization. In *Conference on Learning Theory (COLT)*, 2009.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*, 2017.
- Leon Mirsky. Symmetric gauge functions and unitarily invariant norms. *The Quarterly Journal of Mathematics*, 11(1):50–59, 1960.
- Barak A. Pearlmutter. Fast exact multiplication by the Hessian. *Neural Computation*, 6(1):147–160, 1994.
- Levent Sagun, Utku Evci, V. Uğur Güney, Yann Dauphin, and Léon Bottou. Empirical analysis of the Hessian of over-parametrized neural networks. *arXiv preprint arXiv:1706.04454*, 2017.
- Donald Shepard. A two-dimensional interpolation function for irregularly-spaced data. In *Proceedings of the 23rd ACM National Conference*, pages 517–524, 1968.
- Charles J. Stone. Optimal rates of convergence for nonparametric estimators. *The Annals of Statistics*, 8(6):1348–1360, 1980.
- The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP)*, pages 367–381, 2020.
- Philippe Tillet, Hsiang-Tsung Kung, and David Cox. Triton: An intermediate language and compiler for tiled neural network computations. In *Proceedings of the 3rd ACM SIGPLAN International Workshop on Machine Learning and Programming Languages*, pages 10–19, 2019.
- Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Holger Wendland. Piecewise polynomial, positive definite and compactly supported radial functions of minimal degree. *Advances in Computational Mathematics*, 4(1):389–396, 1995.
- Zhewei Yao, Amir Gholami, Kurt Keutzer, and Michael W. Mahoney. PyHessian: Neural networks through the lens of the Hessian. In *IEEE International Conference on Big Data*, 2020.

A Reproduction

- Hardware: 2× Quadro RTX 6000 (sm_75, 23.5 GiB, 72 SMs), 36 CPU cores.
- Software: PyTorch 2.7.1+cu118 (CUDA 11.8), Python 3.11.14, cuDNN 90100.
- Numerical policy selected at run time: reduction accumulator `float32` (fp64 is rate-limited on this part, so compensated fp32 is used), trajectory storage `float16`, TF32 unavailable, bf16 tensor cores unavailable.
- Every result file records the full environment and the fingerprint of the evaluation set it used.
- Reproduce with `make all`; the Lean proofs with `cd proofs/StamCert && lake build`.